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# GPT Image Generation Models Prompting Guide

This recipe is archived and may reference outdated models or APIs.



Mandeep Singh (OpenAI) , Emre Okcular (OpenAI) , Tiger He (OpenAI)

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## 1. Introduction

OpenAI's gpt-image generation models are designed for production-quality visuals and highly controllable creative workflows. They are well-suited for both professional design tasks and iterative content creation, and support both high-quality rendering and lower-latency use cases depending on the workflow.

Key Capabilities include:

- **High-fidelity photorealism** with natural lighting, accurate materials, and rich color rendering
- **Flexible quality-latency tradeoffs**, allowing faster generation at lower settings while still exceeding the visual quality of prior-generation image models
- **Robust facial and identity preservation** for edits, character consistency, and multi-step workflows
- **Transparent-background PNG and WebP assets (preview)** with `gpt-image-2` for reusable logos, product cutouts, stickers, and presentation graphics
- **Reliable text rendering** with crisp lettering, consistent layout, and strong contrast inside images
- **Complex structured visuals**, including infographics, diagrams, and multi-panel compositions
- **Precise style control and style transfer** with minimal prompting, supporting everything from branded design systems to fine-art styles

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| Model                      | Quality           | input_fidelity                                                                                                        | Resolutions                                                                                  | Recommended use                                                                                                                                                                                                                          |
|----------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>gpt-image-2</code>   | low, medium, high | Disabled. <code>input_fidelity</code> does not work for this model because output is already high fidelity by default | Any resolution that satisfies the constraints below                                          | Recommended default for new builds. Use for highest-quality generation and editing, text-heavy images, photorealism, compositing, identity-sensitive edits, and workflows where fewer retries matter more than the lowest possible cost. |
| <code>gpt-image-1.5</code> | low, medium       | <code>low</code> , <code>high</code>                                                                                  | <code>1024x1024</code> , <code>1024x1536</code> , <code>1536x1024</code> , <code>auto</code> | Keep for existing validated workflows during migration. For new work, prefer <code>gpt-image-2</code> , especially when                                                                                                                  |

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edit an image with a transparent background:

- Set `background="transparent"`.
- Set `output_format="png"` (the default) or `output_format="webp"`. Both support transparency; `jpeg` does not.
- Omit `output_compression` for PNG output. WebP supports optional compression.
- Explicitly request an isolated subject on a fully transparent background, with no scenery, solid backdrop, checkerboard, or unwanted shadows.
- For edits, explicitly preserve the transparent background in each prompt so later steps do not introduce a new background.
- Keep the returned image as a PNG or WebP file and preserve its alpha channel throughout downstream processing.

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|          |                        |                                                                                                                                                                           |
|----------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2K / QHD | <code>2560x1440</code> | Popular widescreen format and recommended upper reliability boundary for <code>gpt-image-2</code>                                                                         |
| 4K / UHD | <code>3840x2160</code> | Experimental upper-end target. If the max-edge rule is enforced literally as <code>&lt; 3840</code> , round down to the nearest valid size such as <code>3824x2144</code> |

When to use which model

- Choose `gpt-image-2` as the default for most production workflows. It is the strongest overall model and the right upgrade target for teams currently using `gpt-image-1.5` or `gpt-image-1` for high-quality outputs.

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models. They are based on patterns that showed up repeatedly in alpha testing across generation, edits, infographics, ads, human images, UI mockups, and compositing workflows.

- **Structure + goal:** Write prompts in a consistent order (background/scene → subject → key details → constraints) and include the intended use (ad, UI mock, infographic) to set the “mode” and level of polish. For complex requests, use short labeled segments or line breaks instead of one long paragraph.
- **Prompt format:** Use the format that is easiest to maintain. Minimal prompts, descriptive paragraphs, JSON-like structures, instruction-style prompts, and tag-based prompts can all work well as long as the intent and constraints are clear. For production systems, prioritize a skimmable template over clever prompt syntax.

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proportion, action geometry, and gaze alignment.

- **Constraints (what to change vs preserve):** State exclusions and invariants explicitly (e.g., “no watermark,” “no extra text,” “no logos/trademarks,” “preserve identity/geometry/layout/brand elements”). For edits, use “change only X” + “keep everything else the same,” and repeat the preserve list on each iteration to reduce drift. If the edit should be surgical, also say not to alter saturation, contrast, layout, arrows, labels, camera angle, or surrounding objects.
- **Transparent backgrounds:** For `gpt-image-2`, transparency is available in preview. Pair `background="transparent"` with `output_format="png"` (the default) or `output_format="webp"`; `jpeg` does not support transparency. Describe the subject as isolated on a fully transparent background and explicitly exclude scenery, solid backdrops, checkerboards, and unwanted shadows. For edits, repeat “preserve the transparent background” so the workflow does not

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Put any reference images used for edits into `input_images/` (or update the paths in the examples). When saving transparent outputs, use a `.png` or `.webp` extension matching the selected output format and preserve the returned image bytes; converting an RGBA image to RGB discards its transparent background.

```
1 import os
2 import base64
3 from openai import OpenAI
4
5 client = OpenAI()
6
7 os.makedirs("../..//images/input_images", exist_ok=True)
8 os.makedirs("../..//images/output_images", exist_ok=True)
9
10 def save_image(result, filename: str) -> None:
```



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## 4. Use Cases — Generate (text → image)

### 4.1 Infographics

Use infographics to explain structured information for a specific audience: students, executives, customers, or the general public. Examples include explainers, posters, labeled diagrams, timelines, and “visual wiki” assets. For dense layouts or heavy in-image text, it’s recommended to set output generation quality to “high”.

```
1 prompt = ""  
2 Create a detailed Infographic of the functioning and flow  
  of an automatic coffee machine like a Jura.  
3 From bean basket, to grinding, to scale, water tank,  
  boiler, etc.  
4 I'd like to understand technically and visually the flow.  
5 ""
```



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## 4.2 Translation in Images

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### 4.3 Photorealistic Images that Feel “natural”

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Output Image:

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#### 4.4 World knowledge

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## 4.5 Logo Generation

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```
14     output_format= 'png' ,
15     n=4,      # Generate 4 versions of the logo
16 )
17
18 # Save all 4 images to separate files
19 for i, item in enumerate(result.data, start=1):
20     image_base64 = item.b64_json
21     image_bytes = base64.b64decode(image_base64)
22     with
23         open(f"../../images/output_images/logo_generation_{i}_gpt-
24             image-2.png", "wb") as f:
25             f.write(image_bytes)
```

Output Images:

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```
1 prompt = ""
2 Give me a cool in culture ad / fashion shot for a brand
  called Thread.
3 It's a hip young street brand. The ad shows a group of
  friends hanging out together with the tagline "Yours to
  Create."
4 Make it feel like a polished campaign image for a youth
  streetwear audience: stylish, contemporary, energetic, and
  tasteful.
5 Use clean composition, strong color direction, natural
  poses, and premium fashion photography cues.
6 Render the tagline exactly once, clearly and legibly,
  integrated into the ad layout.
7 No extra text, no watermarks, no unrelated logos.
8 ""
9
10 result = client.images.generate()
```



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## 4.7 Story-to-Comic Strip

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Output Image:

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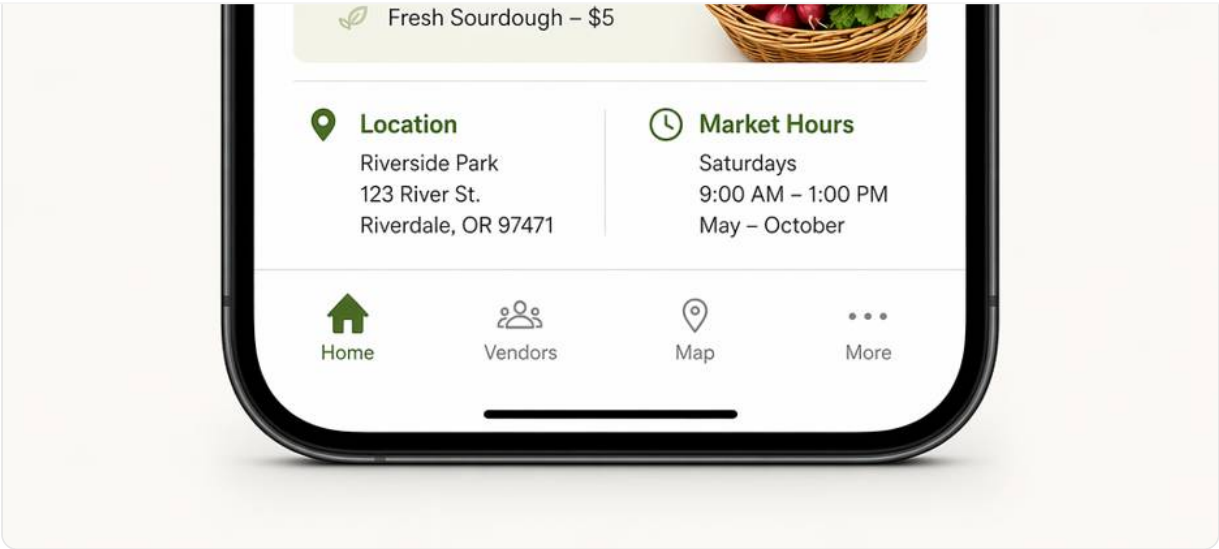


## 4.8 UI Mockups

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Output image.

Loading docs agent...



4.9 Scientific / Educational Visuals

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```
11 result = client.images.generate(  
12     model="gpt-image-2",  
13     prompt=prompt,  
14     size="1536x1024",  
15     quality="high",  
16 )  
17  
18 save_image(result,  
    "scientific_educational_cellular_respiration_gpt-image-  
    2.png")
```

Output Image:

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For slides, charts, and diagram-heavy assets, include the numbers and labels directly in the prompt. Use a landscape size for deck-style outputs and `quality="high"` when the image contains small text, legends, axes, or footnotes.

```
1 prompt = ""
2 Create one pitch-deck slide titled "Market Opportunity"
  that feels like a real Series A fundraising slide from a
  YC-backed startup.
3
4 Use a clean white background, modern sans-serif typography
  like Inter, and a crisp, minimal layout. The slide should
  include:
5
6 * A TAM/SAM/SOM concentric-circle diagram in muted blues
  and grays
7 * Specific, believable market sizing numbers:
```



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# Market Opportunity

Huge market. Early innings. Clear wedge.

**TAM**  
**\$42B**

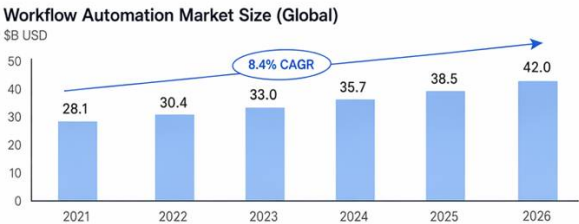
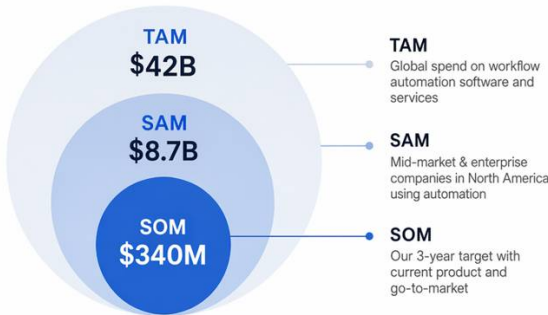
**Total Addressable Market**  
Global spend on workflow automation software and services across knowledge work industries.

**SAM**  
**\$8.7B**

**Serviceable Addressable Market**  
Mid-market and enterprise companies in North America using automation to improve GTM and back-office workflows.

**SOM**  
**\$340M**

**Serviceable Obtainable Market**  
Our realistic 3-year target capturing a portion of the SAM with our initial product and go-to-market motion.



1. AGI Research, 2024  
2. Internal analysis

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Output Image:

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## 5.2 Virtual Clothing Try-On

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```
    "rb"),
12         open("../..//images/input_images/boots.png", "rb"),
13     ],
14     prompt=prompt,
15     size="1024x1536",
16     quality="medium",
17 )
18
19 save_image(result, "outfit_gpt-image-2.png")
```

Input Images:

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Loading docs agent...

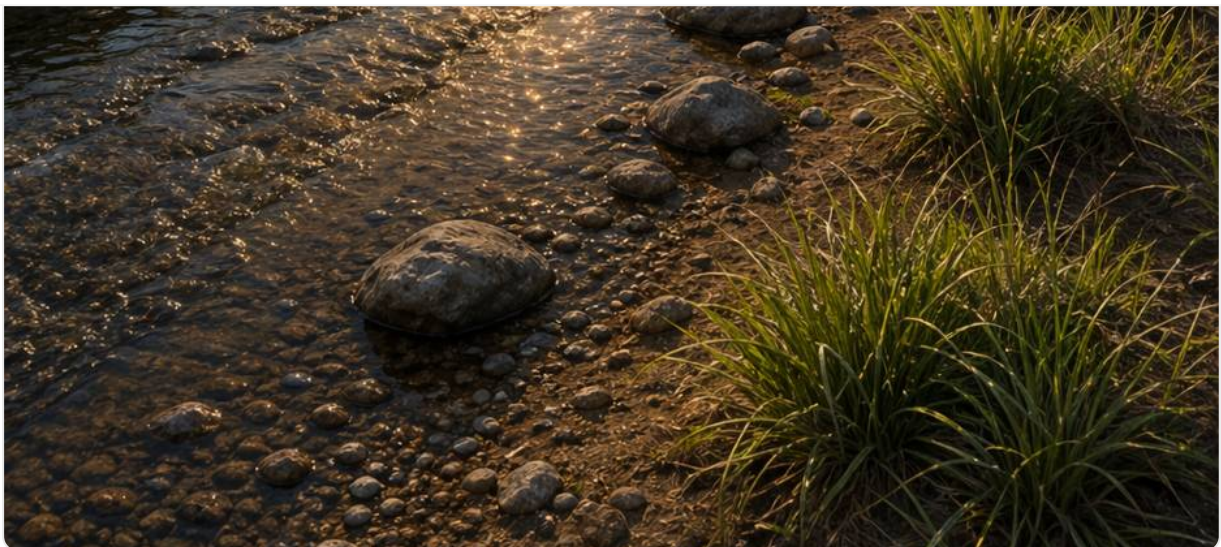


### 5.3 Drawing → Image (Rendering)

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#### 5.4 Product Mockups (transparent background + label integrity)

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```
13     ],
14     prompt=prompt,
15     size="1024x1536",
16     quality="medium",
17     background="transparent",
18     output_format="png",
19 )
20
21 save_image(result, "extract_product_gpt-image-2.png")
```

Input Image:

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12 FL OZ (355 mL)

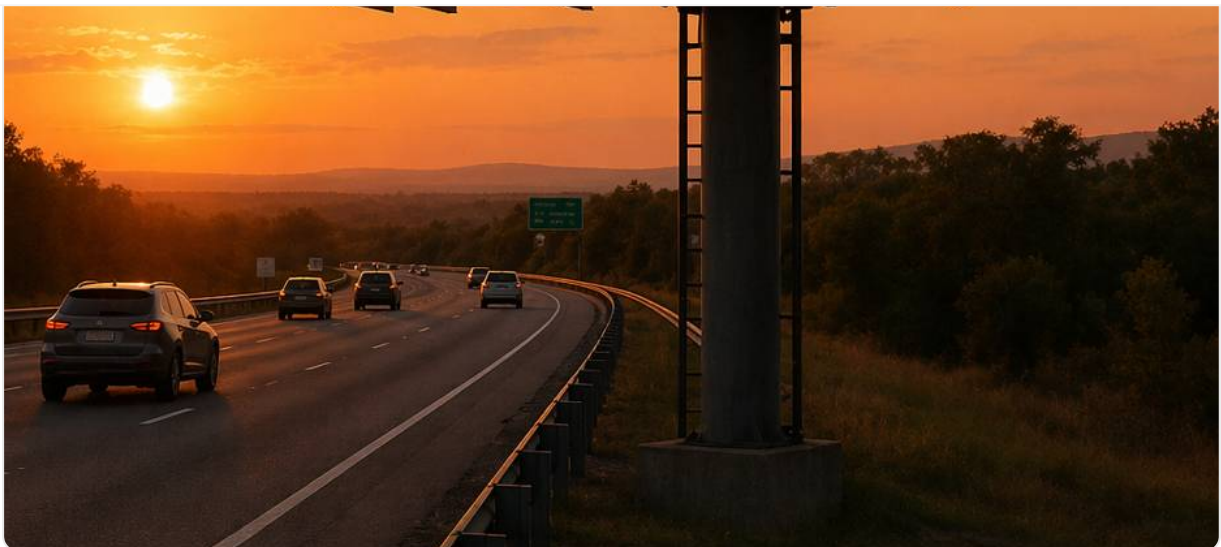
## 5.5 Marketing Creatives with Real Text In-Image

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Input Image:

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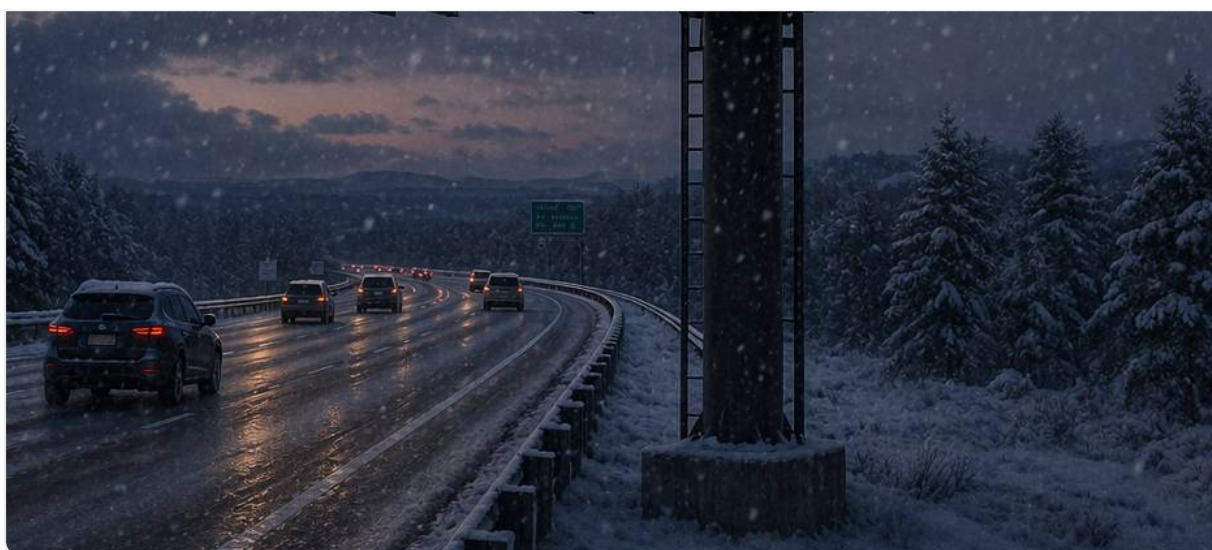
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## 5.6 Lighting and Weather Transformation

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## 5.7 Object Removal

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scenarios where facial/identity preservation matters. Anchor realism by specifying a grounded photographic look (natural lighting, believable detail, no cinematic grading), and lock what must not change about the subject. When available, higher input fidelity helps maintain likeness during larger scene edits.

```
1 prompt = ""  
2 Generate a highly realistic action scene where this person  
   is running away from a large, realistic brown bear  
   attacking a campsite. The image should look like a real  
   photograph someone could have taken, not an overly enhanced  
   or cinematic movie-poster image.  
3 She is centered in the image but looking away from the  
   camera, wearing outdoorsy camping attire, with dirt on her  
   face and tears in her clothing. She is clearly afraid but  
   focused on escaping, running away from the bear as it  
   destroys the campsite behind her.
```



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## 5.9 Multi-Image Referencing and Compositing

Used to combine elements from multiple inputs into a single, believable image—great for “insert this object/person into that scene” workflows without re-generating everything. The key is to clearly specify what to transplant (the dog from image 2), where it should go (right next to the woman in image 1), and what must remain unchanged (scene, background, framing), while matching lighting, perspective, scale, and shadows so the composite looks naturally captured in the original photo.

```
1 prompt = ""  
2 Place the dog from the second image into the setting of  
  image 1, right next to the woman, use the same style of  
  lighting, composition and background. Do not change  
  anything else.  
3 ""  
4
```



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## 6. Additional High-Value Use Cases

### 6.1 Interior design “swap” (precision edits)

Used for visualizing furniture or decor changes in real spaces without re-rendering the entire scene. The goal is surgical realism: swap a single object while preserving

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## 6.2 3D pop-up holiday card (product-style mock)

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```
24 - Original artwork only
25 - No trademarks
26 - No watermarks
27 - No logos
28
29 Include ONLY this card text (verbatim):
30 "{short_copy}"
31 ""
32
33 result = client.images.generate(
34     model="gpt-image-2",
35     prompt=prompt,
36     size="1024x1536",
37     quality="medium",
38 )
39
40 save_image(result, "christmas_holiday_card_teddy_gpt-image-2.png")
```

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```
19 style:
20   Premium toy photography, realistic plastic and painted
21   metal textures,
22   studio lighting, shallow depth of field,
23   sharp label printing, high-end retail presentation.
24 Constraints:
25   - Original design only
26   - No trademarks
27   - No watermarks
28   - No logos
29
30 Include ONLY this packaging text (verbatim):
31 "{short_copy}"
32 ""
33
34 result = client.images.generate(
35     model="gpt-image-2",
36     prompt=prompt,
```

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## 6.4 Children's Book Art with Character Consistency (multi-image workflow)

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```
17 Proportions suitable for picture books (slightly oversized
    head, expressive face).
18
19 Constraints:
20 - Original character (no copyrighted characters)
21 - No text
22 - No watermarks
23 - Plain forest background to clearly showcase the character
24 """"
25
26 # ---- Image generation ----
27 result = client.images.generate(
28     model="gpt-image-2",
29     prompt=prompt,
30     size="1024x1536",
31     quality="medium",
32 )
33
```

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2 Story continuation — reuse character, advance the narrative

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```
26 result = client.images.edit(  
27     model="gpt-image-2",  
28     image=[  
29         open("../images/output_images/childrens_book_illustratio  
n_1_gpt-image-2.png", "rb"), # use image from step 1  
30     ],  
31     prompt=prompt,  
32     size="1024x1536",  
33     quality="medium",  
34 )  
35  
36 save_image(result, "childrens_book_illustration_2_gpt-  
image-2.png")
```

Output Image:

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**Conclusion**

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